

# **Multiplicative Information Gating: Decomposing News into Novelty, Materiality and Polarity for Calibrated Selective Forecasting**

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## **Abstract**

Systems that turn text into decisions usually reduce each event to one sentiment polarity and sum, conflating three properties that move prices through different channels. We propose multiplicative information gating: an event is decomposed into polarity, novelty and materiality and combined as their product, so any near-zero factor vetoes the signal. We prove the product is the canonical combiner under per-axis scaling and bound the signal by its weakest factor. A language model supplies the three scores; the gated signal feeds a forecaster that is calibrated and then permitted to abstain. On a live ledger of 621 resolved forecasts, nominal 90% intervals covered 70.7%. An offline three-stage recalibration cut expected calibration error from 0.35 to 0.05. Single-name direction proved unpredictable beyond drift at 50.1%, so a conviction gate abstaining on 94% of cases raised the thirty-day hit rate to 60.6% against a 58.0% base.

**Keywords:** Calibration; Selective prediction; Sentiment analysis;  
Large language model; Financial forecasting

## 1. Introduction

A growing class of decision systems reads text and acts on it. The dominant recipe scores each item for sentiment polarity, averages the scores and feeds the average to a predictor [1, 2], an approach that has shown media tone to carry return-relevant information [1, 3]. Polarity answers only one question, though: is this good or bad? It says nothing about whether the news is new or a restatement of something priced weeks ago, nor whether it bears on value or is merely cosmetic, so summed, loud and empty items drown out the rare informative ones.

We therefore take a textual event to have three separable properties, each governing price impact through a different channel: *polarity*, the sign of the implied value change; *novelty*, how much genuinely new information it carries; and *materiality*, how sensitive fundamental value is to the event, independent of sign or freshness. Multiplicative information gating combines them as a product, which behaves like a soft logical AND: a stale or immaterial item yields a signal near zero however strong its polarity, so one weak factor vetoes the event.

Who supplies the scores? Dictionaries capture polarity passably [2] but struggle with novelty and materiality, which require world knowledge; we use an instruction-tuned large language model prompted with numerical anchors [4, 5]. And how do we keep the forecaster honest? One near the efficient-market limit [6] is mostly wrong about direction, so it owes the user a calibrated probability and the discipline to abstain when conviction is low [7, 8]. Our objectives are to formalise the decomposition and its properties (Sections 2.1 and 3), to give a reproducible scoring protocol (Section 2.2), and to couple the gated signal to a calibrated selective forecaster, reporting measured results only (Section 4).

## 1.1 Related work

Tetlock [1] linked pessimistic media tone to downward pressure and reversal, Antweiler and Frank [3] studied message boards, and Loughran and McDonald [2] showed finance-specific dictionaries matter because general-purpose ones mislabel domain terms. Supervised text models improve on dictionaries [9] and pre-trained language models now dominate sentiment extraction [10], but all treat the per-item output as a scalar polarity, handling novelty and materiality by ad hoc filters if at all. Boudoukh et al. [11] is an exception in spirit: identifying which news is relevant sharply changes the measured information content.

Instruction-tuned models emit structured numeric judgements from rubric-style prompts [4] and finance has adopted them quickly [5], though their self-reported confidence is imperfectly calibrated [12]. We do not ask the model to forecast but to *decompose*, a narrower and more checkable task; to our knowledge no prior work scores novelty and materiality as separate, continuous per-item quantities and makes them multiplicative preconditions for influence with a provable veto, existing pipelines [2, 5, 9, 10] reducing the item to one polarity first. Modern classifiers are often miscalibrated [13]; remedies include Platt scaling [14] and isotonic regression [15], assessed by expected calibration error and the Brier score [16], while stacked generalisation [17] complicates this, a failure we repair in Section 2.3. Abstention is an old idea [7] with a modern theory [8] and a distribution-free cousin in conformal prediction [18]. Financial machine learning is dominated by accuracy and area-under-the-curve reporting, fragile under backtest overfitting [19]; deep and ensemble models are standard [20], but few report whether their probabilities are trustworthy.

## 2. Materials and Methods

### 2.1 The multiplicative information model

Let  $e$  denote a textual event (a news headline) about an entity (a listed equity), with scores  $s(e) \in [-1, 1]$  (polarity),  $\nu(e) \in [0, 1]$  (novelty, surprise relative to the market's prior) and  $\mu(e) \in [0, 1]$  (materiality, the sensitivity of value to the event regardless of sign). The *event signal* is their product,

$$a(e) = s(e) \nu(e) \mu(e) \in [-1, 1]. \quad (1)$$

Define *relevance*  $r(e) = \nu(e) \mu(e)$ , so  $a(e) = s(e) r(e)$ . To aggregate the events  $\mathcal{E}$  touching one entity we use a relevance-weighted mean restricted to events clearing a materiality floor  $\mu_0$ ,

$$A = \frac{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e) a(e)}{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e)} = \frac{\sum r(e)^2 s(e)}{\sum r(e)}. \quad (2)$$

Relevance enters twice, inside the event signal and again as the weight, so a novel and material item dominates quadratically while cosmetic chatter is suppressed; we use  $\mu_0 = 0.15$ . To first order the expected move is the product of a sign, a sensitivity and a surprise magnitude; additive sentiment keeps only the sign, which is why it over-reacts to loud, stale or immaterial items.

### 2.2 Scoring the three axes with a language model

The scores come from an instruction-tuned large language model (in our deployment, Claude Haiku 4.5) prompted as a quantitative analyst and constrained to return a JSON array of  $\{\nu, \mu, s\}$  triples, one per headline, in input order. Two choices make them reproducible. First, the prompt fixes *numerical anchors* per axis: for novelty, 0.0 is an exact restatement of news priced weeks ago, 0.2 an in-line result, 0.6 an unexpected but modest development and 1.0 a sudden regulatory ban; for materiality, 0.0 is boilerplate,

0.4 a contract worth a few percent of revenue, 0.8 a transformative deal and 1.0 an existential event; for polarity, anchors run from  $-1.0$  (fraud allegation) through  $0.0$  (a neutral reshuffle) to  $+1.0$  (a large earnings beat). Anchoring makes two runs on the same item agree closely, which matters because the product in Eq. (1) amplifies disagreement on any one axis. Second, every scored item is memoised in a content-addressed store keyed by a hash of the entity and normalised headline, so a run is deterministic. When the model is unavailable, a lexicon fallback assigns polarity from curated term lists with fixed conservative novelty and materiality, flagged for downstream discounting.

### 2.3 From gated signal to calibrated probability

The aggregate  $A$  of Eq. (2) is one feature among several entering a hybrid forecaster blending a gradient-boosted tree ensemble with a recurrent sequence model over price, momentum, trend, reversal and volatility features, producing per name and horizon a point forecast, a price interval and a directional probability  $\hat{p}$ . We treat it as a black box and study its calibration, measured by the expected calibration error over  $B$  equal-width bins,

$$\text{ECE} = \sum_{b=1}^B \frac{|G_b|}{N} |\bar{y}_b - \bar{p}_b|, \quad (3)$$

with  $\bar{p}_b, \bar{y}_b$  the mean predicted probability and mean realised outcome in bin  $b$ , and by the Brier score  $\frac{1}{N} \sum_i (\hat{p}_i - y_i)^2$  [16]. Our first deployed model reported a next-day up-probability near 0.80 against a realised accuracy near 0.51. The cause was a stacking domain mismatch [17]: an isotonic calibrator fitted on the out-of-fold *average of the base learners* was applied to the *stacked* prediction, a different distribution, and the blend was never itself calibrated. The repair is a three-stage calibration: refit the isotonic map on the base-average it was meant to correct, form the blend, then fit a final Platt scaler [14] on a held-out fold and apply it to the test and live rows.

## 2.4 The conviction gate: abstention as decision-stage gating

Calibration makes  $\hat{p}$  honest, not skilful. If short-horizon direction is near-random (Section 4.3), a calibrated probability sits near the base rate almost everywhere and acting on every name is pointless. The remedy mirrors the text-stage veto: act only when conviction clears a threshold. For a horizon of  $H$  trading days we emit an UP call when  $\hat{p} \geq \tau^*$  and NEUTRAL otherwise, choosing  $\tau^*$  on training folds as the smallest  $\tau$  whose fired bucket reaches a target precision (0.60) while still firing at least a minimum fraction of the time (5%):

$$\tau^* = \min \left\{ \tau : \text{prec}(\tau) \geq 0.60 \text{ and } \phi(\tau) \geq 0.05 \right\}, \quad (4)$$

where  $\text{prec}(\tau) = \Pr[y = 1 \mid \hat{p} \geq \tau]$  is the fired-bucket precision and  $\phi(\tau) = \Pr[\hat{p} \geq \tau]$  the firing rate. The gate is intentionally one-sided: confident DOWN calls are destroyed by the upward drift of equities (Section 4.4), so the system emits long-or-neutral signals only. Like the event veto, it forgoes breadth for dependability, so the quantity worth reporting is precision at a given firing rate, not accuracy or area under the curve.

## 2.5 Data, deployment and evaluation protocol

The system is deployed on liquid National Stock Exchange of India (NSE) large-capitalisation equities, and two bodies of evidence are kept strictly separate. The first is an *offline walk-forward* study on daily data from 2018 to 2026 for a 54-name universe; all offline statistics are out-of-sample under an expanding window, so no future information enters a fitted fold. The second is a *live append-only ledger*: since April 2026 the system has written every issued forecast (entity, timestamp, target date, point forecast, interval, nominal confidence, directional probability, horizon) to a row later stamped with the realised price and outcome, scored first-look and never retro-edited.

### 3. Theory

The multiplicative form carries a guarantee no additive combiner offers.

**Proposition 1** (Veto bound). *For every event  $e$ ,  $|a(e)| \leq \min\{|s(e)|, \nu(e), \mu(e)\}$ . In particular  $a(e) = 0$  whenever any single factor is zero, and the aggregate satisfies  $|A| \leq \max_e |s(e)|$ .*

*Proof.* Each factor lies in  $[-1, 1]$  with  $\nu, \mu \geq 0$ , so  $|a(e)| = |s(e)| \nu(e) \mu(e)$  is a product of three numbers of modulus at most one; such a product cannot exceed the modulus of any one of them, which gives the per-event bound and the zero case. For the aggregate, Eq. (2) gives  $A = (\sum_e r(e)^2 s(e)) / (\sum_e r(e))$ , so

$$|A| \leq \frac{\sum_e r(e)^2 |s(e)|}{\sum_e r(e)} \leq \max_e |s(e)| \cdot \frac{\sum_e r(e)^2}{\sum_e r(e)} \leq \max_e |s(e)|,$$

where the last step uses  $\sum_e r(e)^2 \leq (\max_e r(e)) \sum_e r(e) \leq \sum_e r(e)$  because every  $r(e) \in [0, 1]$ . The weights are non-negative but sub-stochastic, so the aggregate contracts the polarities toward zero and never amplifies them.  $\square$

Proposition 1 formalises “a single weak factor vetoes the event”: a stale ( $\nu \rightarrow 0$ ) or immaterial ( $\mu \rightarrow 0$ ) item is capped near zero however extreme its polarity. No additive combination does this, since a weighted sum  $\alpha s + \beta \nu + \gamma \mu$  is generally nonzero when one term vanishes.

**Proposition 2** (Characterisation). *Consider combiners  $f(s, \nu, \mu)$  with  $s \in [-1, 1]$ ,  $\nu, \mu \in [0, 1]$  satisfying: (A0) normalisation,  $f(1, 1, 1) = 1$ ; (A1) sign fidelity,  $\text{sign } f = \text{sign } s$ ; (A2) veto,  $f = 0$  if any of  $|s|, \nu, \mu$  is 0; (A3) separable monotonicity,  $|f|$  is non-decreasing in each of  $\nu$  and  $\mu$  for fixed others, and  $f$  is non-decreasing in  $s$  for fixed  $\nu, \mu \geq 0$ ; and (A4) factor homogeneity, scaling any single argument by  $\lambda \in [0, 1]$  scales  $f$  by  $\lambda$ . Then  $f(s, \nu, \mu) = s \nu \mu$ .*

*Proof sketch.* By (A4) applied to each argument in turn, scaling the arguments by  $\lambda_1, \lambda_2, \lambda_3$  in  $[0, 1]$  scales  $f$  by  $\lambda_1 \lambda_2 \lambda_3$ . Writing  $s = \text{sign}(s) |s|$  and taking  $\lambda_1 = |s|, \lambda_2 = \nu, \lambda_3 = \mu$  relative to the unit corner gives  $f(s, \nu, \mu) = |s| \nu \mu f(\text{sign}(s), 1, 1)$ . Sign fidelity (A1) forces  $f(\pm 1, 1, 1) = \pm c$  with  $c = f(1, 1, 1)$ , and normalisation (A0) fixes  $c = 1$ . Hence  $f = s \nu \mu$ ; (A2) and (A3) are then automatically satisfied, so they are consistency requirements rather than the active constraints.  $\square$

The force of the result lies in homogeneity (A4): it requires each axis to act on its own multiplicative scale, and once that and (A0) are granted the product follows.

Since (A2) and (A3) do not by themselves single out the product, we claim only that it is the canonical combiner once per-axis scaling is imposed — exactly what a position-taking system needs to treat freshness and relevance as preconditions for influence.

## 4. Results and Discussion

### 4.1 Deployed uncertainty does not self-correct

Between 19 April and 12 June 2026 the system issued and later resolved 621 price-interval forecasts on seven NSE large caps (ABB, HDFCBANK, ICICIBANK, INFY, RELIANCE, SBIN, TCS), each carrying a nominal 90% interval with a median half-width of 5.6% of price. Had the intervals been well calibrated, about 90% would contain the realised price; empirical coverage was 70.7% overall, a 19.3 percentage-point shortfall. Most forecasts were at the ten-day horizon, where coverage was 69.2% over 575 resolved forecasts; the smaller twenty-day and five-day buckets covered 86.8% and 100% (Table 1a, Figure 1a). Stated confidence was systematically overstated and nothing in the pipeline noticed: the empirical reason calibration and abstention cannot be afterthoughts.

### 4.2 A stacking domain mismatch, and a three-stage repair

This and the two results that follow are walk-forward measurements, not live-ledger figures. The original next-day directional head reported probabilities clustered near 0.80 against a realised accuracy of about 0.51, giving  $ECE = 0.345$  and a Brier score above 0.366, worse than always predicting 0.5. The three-stage repair of Section 2.3 cut ECE from 0.345 to 0.049 and pulled the reported up-probability back to a realistic neighbourhood of 0.50. Accuracy did not move: calibration changed what the model *said about its confidence*, not what it predicted. A calibrator is bound to the distribution it was fitted on, and stacking creates more than one.



### 4.3 Single-name direction is unpredictable beyond drift

With probabilities pulled back to the base rate, we asked whether any single-name directional edge survives at short horizons. A cross-sectional model predicting five-day direction reached 50.09% out-of-fold accuracy, with a ranking area under the curve near 0.49; that is chance. We confirmed this five ways, varying the learner and feature set and gating by trend and market regime; none produced a stable edge over the always-up base rate. This matches the efficient-market prediction for liquid names [6] and reframes the problem: if ordering names by predicted return is near-random, the only lever left is *selection*.

### 4.4 Selection recovers a small, stable edge

Lengthening the horizon to thirty trading days opens two modest edges: the upward drift of equities (about 58% of thirty-day windows are up) and a small idiosyncratic component captured by a calibrated cross-sectional model on momentum, trend, reversal and volatility features. Walk-forward from 2018 to 2026 over the 54-name universe (91,471 training rows, 50,272 out-of-sample rows), the gate of Eq. (4) settles at  $\tau^* = 0.63$ . The fired UP bucket achieves 60.6% directional precision against a 58.0% always-up base rate, a net edge of 2.6 percentage points, while firing on only 5.6% of observations; the model abstains on the remaining 94%. The bucket pools to 2,840 calls, so the 60.6% precision carries a 95% binomial interval of 58.8 to 62.4%, and a one-sided test against the base gives  $z \approx 2.9$  ( $p \approx 0.002$ ). We do not over-read it: per-year fired counts are small (Table 1b), so the swings from +10.8 points in 2025 to −2.4 in 2024 are within sampling noise, and the honest summary is a small but repeatable tail edge. The ranking area under the curve is about 0.47: the edge lives in the high-conviction tail, not the ordering, which is what precision at a fixed firing rate registers and area under the curve overlooks. The gate beats its base rate in four of five test years (Table 1b, Figure 1b). Confident DOWN calls reach only about 10% precision out-of-sample, destroyed by drift, which is why the gate never emits them.

## 4.5 The veto in action

The text stage cannot be evaluated as an isolated alpha, because the deployed ledger stores no per-event return labels; what we can show is that the decomposition behaves as designed. Scoring four headlines for one entity under the anchors of Section 2.2 gives  $a = +0.018$  for a consensus-meeting profit in an upbeat tone ( $s = +0.30$ ,  $\nu = 0.20$ ,  $\mu = 0.30$ ),  $+0.022$  for a reiterated buyback  $(+0.55, 0.10, 0.40)$  and  $+0.003$  for a routine reshuffle  $(+0.05, 0.40, 0.15)$ , but  $-0.330$  for an unexpected regulator notice on a core product line  $(-0.55, 0.75, 0.80)$ . The glowing but routine item and the emphatic but cosmetic one are gated down an order of magnitude, while the quieter disclosure that is new and material dominates despite its moderate polarity — the protection Proposition 1 guarantees and additive sentiment cannot. These values illustrate scoring behaviour and carry no performance claim.

## 4.6 Discussion, threats to validity and honest limits

Every result turns on one trade-off: the text-stage veto refuses stale or immaterial items and the decision-stage gate stays out on most names, so the system gives up commenting on everything and earns the right to be believed on the little it flags. Nothing in Section 2 is specific to equities: any stream of timestamped textual events driving a decision has the same three axes, and triage of clinical alerts, operations tickets and moderation queues fail the same way when collapsed to one urgency score. The gate is an instance of selective classification [7, 8], paired here with a probability whose calibration was repaired; distribution-free interval methods [18] address the coverage failure of Section 4.1.

Four limits deserve emphasis. The language-model scores are judgements, not measurements: anchoring tightens them but a different model or prompt could shift the scale. The live ledger is young — two months and seven names — so its coverage shortfall is a snapshot, not a stationary law. Weightiest, we have no per-event return labels in deployment, so the marginal value of the decomposition over plain polarity is

argued from its formal properties and its place in the measured pipeline rather than an isolated backtest, which would invite the backtest-overfitting critique [19]. Finally, the live directional-probability sample is small, 28 resolved next-period calls, so our live claim rests on the 621-forecast interval-coverage evidence.

## 5. Conclusions

A textual event carries direction, surprise and consequence, and these need not move together; their product builds a signal whose weakest axis caps the rest, keeping a live decision from chasing loud noise, and it is the canonical combiner once per-axis scaling is imposed. The same logic belongs at the decision stage: nominal 90% intervals covered only 70.7% of live outcomes, a three-stage recalibration cut expected calibration error from 0.35 to 0.05, and a conviction gate abstaining on 94% of names raised the thirty-day hit rate to 60.6% against a 58.0% base.

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**Declaration of generative AI.** An instruction-tuned model (Claude Haiku 4.5, Anthropic) is used as a measurement instrument within the system under study, scoring headlines from the anchored prompt of Section 2.2; those outputs were validated against the deterministic fallback and persisted for audit. During manuscript preparation a large language model was used solely to improve language and readability; no generative tool produced data, results, figures or interpretations, and the authors take full responsibility.

**Data availability.** The live forecast ledger and the offline walk-forward statistics underlying Table 1 and Figure 1 are available from the corresponding author on reasonable request. Price data are public NSE end-of-day records.

## 7. References

- [1] Tetlock, P.C., 2007. Giving content to investor sentiment: the role of media in the stock market. *Journal of Finance*, 62, 1139–1168.
- [2] Loughran, T., McDonald, B., 2011. When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66, 35–65.
- [3] Antweiler, W., Frank, M.Z., 2004. Is all that talk just noise? The information content of internet stock message boards. *Journal of Finance*, 59, 1259–1294.
- [4] Brown, T.B., Mann, B., Ryder, N., et al., 2020. Language models are few-shot learners. In: *Adv. Neural Inf. Process. Syst.* 33, pp. 1877–1901.
- [5] Lopez-Lira, A., Tang, Y., 2023. Can ChatGPT forecast stock price movements? Return predictability and large language models. arXiv:2304.07619. [cited 2026 Aug 5]. Available from: <https://arxiv.org/abs/2304.07619>.
- [6] Fama, E.F., 1970. Efficient capital markets: a review of theory and empirical work. *Journal of Finance*, 25, 383–417.
- [7] Chow, C.K., 1970. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16, 41–46.
- [8] Geifman, Y., El-Yaniv, R., 2017. Selective classification for deep neural networks. In: *Adv. Neural Inf. Process. Syst.* 30, pp. 4878–4887.
- [9] Ke, Z.T., Kelly, B.T., Xiu, D., 2019. Predicting returns with text data. National Bureau of Economic Research Working Paper 26186, Cambridge.
- [10] Huang, A.H., Wang, H., Yang, Y., 2023. FinBERT: a large language model for extracting information from financial text. *Contemporary Accounting Research*, 40, 806–841.
- [11] Boudoukh, J., Feldman, R., Kogan, S., Richardson, M., 2019. Information, trading, and volatility: evidence from firm-specific news. *Review of Financial Studies*, 32, 992–1033.
- [12] Steyvers, M., Tejeda, H., Kumar, A., et al., 2025. What large language models know and what people think they know. *Nature Machine Intelligence*, 7, 221–231.
- [13] Guo, C., Pleiss, G., Sun, Y., Weinberger, K.Q., 2017. On calibration of modern neural networks. In: *Proc. 34th Int. Conf. Mach. Learn.*, pp. 1321–1330.
- [14] Platt, J.C., 1999. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In: *Adv. Large Margin Classifiers*. MIT Press, pp. 61–74.
- [15] Zadrozny, B., Elkan, C., 2002. Transforming classifier scores into accurate multi-class probability estimates. In: *Proc. 8th ACM SIGKDD Int. Conf.*, pp. 694–699.
- [16] Brier, G.W., 1950. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78, 1–3.
- [17] Wolpert, D.H., 1992. Stacked generalization. *Neural Networks*, 5, 241–259.
- [18] Angelopoulos, A.N., Bates, S., 2023. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *Foundations and Trends in Machine Learning*, 16, 494–591.
- [19] Bailey, D.H., Borwein, J., Lopez de Prado, M., Zhu, Q.J., 2017. The probability of backtest overfitting. *Journal of Computational Finance*, 20, 39–69.
- [20] Gu, S., Kelly, B., Xiu, D., 2020. Empirical asset pricing via machine learning. *Review of Financial Studies*, 33, 2223–2273.

## Figure Captions

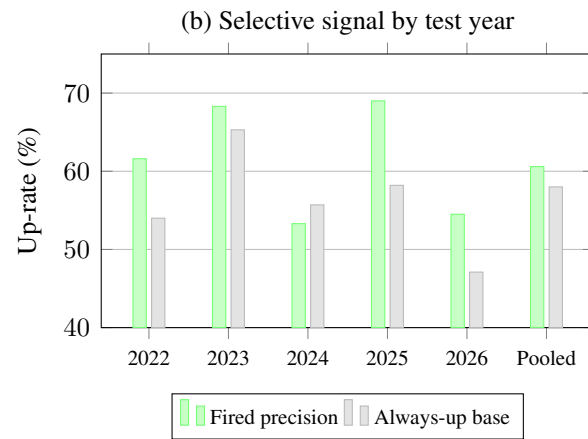
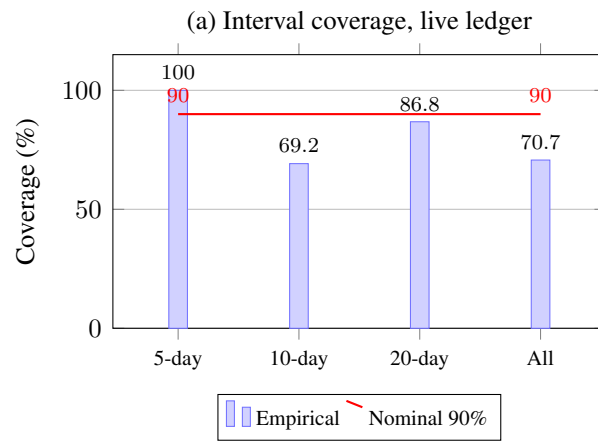
**Figure 1.** Measured behaviour of the two dependability mechanisms. (a) Empirical coverage of nominal 90% intervals on the live ledger by horizon (resolved forecasts: 8, 575, 38, 621 in total). (b) Fired-bucket precision of the conviction gate against the always-up base rate by test year, at about a 5.6% firing rate.

**Table 1.** Measured results. Panel (a): live-ledger interval coverage by horizon (NSE large caps, 19 April–12 June 2026; every interval carries a nominal 90% level). Panel (b): walk-forward selective signal by test year, where precision is the realised up-rate of the fired high-conviction bucket, base is the always-up rate and fires is the number of high-conviction calls.

| <b>(a) Live-ledger interval coverage by horizon</b> |                           |                |                  |
|-----------------------------------------------------|---------------------------|----------------|------------------|
| <b>Horizon</b>                                      | <b>Resolved forecasts</b> | <b>Nominal</b> | <b>Empirical</b> |
| 5 trading days                                      | 8                         | 90.0%          | 100.0%           |
| 10 trading days                                     | 575                       | 90.0%          | 69.2%            |
| 20 trading days                                     | 38                        | 90.0%          | 86.8%            |
| <b>All</b>                                          | <b>621</b>                | <b>90.0%</b>   | <b>70.7%</b>     |

| <b>(b) Walk-forward selective signal by test year</b> |              |                        |                          |                  |
|-------------------------------------------------------|--------------|------------------------|--------------------------|------------------|
| <b>Test year</b>                                      | <b>Fires</b> | <b>Fired precision</b> | <b>Always-up base</b>    | <b>Edge (pp)</b> |
| 2022                                                  | 365          | 61.6%                  | 54.0%                    | +7.6             |
| 2023                                                  | 290          | 68.3%                  | 65.3%                    | +3.0             |
| 2024                                                  | 932          | 53.3%                  | 55.7%                    | −2.4             |
| 2025                                                  | 822          | 69.0%                  | 58.2%                    | +10.8            |
| 2026                                                  | 431          | 54.5%                  | 47.1%                    | +7.4             |
| <b>Pooled</b>                                         | <b>2,840</b> | <b>60.6%</b>           | <b>58.0%<sup>†</sup></b> | <b>+2.6</b>      |

<sup>†</sup>The pooled base is the unconditional always-up rate over all out-of-sample rows, not the fires-weighted mean of the yearly bases (55.9%); the pooled edge is therefore the conservative of the two comparisons.



**Figure 1.** Measured behaviour of the two dependability mechanisms (see Figure Captions).